

Test for difference of means

Suppose if we want to test, if two independent samples have been drawn from two normal population having the same mean and population variance be equal.

Let $x_1, x_2, \dots, x_n$ & $y_1, y_2, \dots, y_n$ be two independent random samples from the given normal population, we setup the null hypothesis $H_0: \mu_x = \mu_y$.

i.e., the samples have been drawn from normal population with the same mean.

In other words, sample means $\bar{x}$ & $\bar{y}$ do not differ significantly under the assumption that $\sigma_1^2 = \sigma_2^2 = \sigma^2$

i.e., population variances are equal but unknown.

Test statistic, $t = \frac{\bar{x} - \bar{y}}{S \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$ where

$$S^2 = \frac{1}{n_1 + n_2 - 2} \left[\sum (x_i - \bar{x})^2 + \sum (y_i - \bar{y})^2 \right]$$

and $\bar{x} = \frac{\sum x_i}{n}$, $\bar{y} = \frac{\sum y_i}{n}$

follows a 't' distribution with $(n_1 + n_2 - 2)$ degrees of freedom.

Ques. Sample A = 24, 27, 26, 21, 25

Sample B = 27, 30, 28, 31, 22, 36

Can it be said that 2 samples came from normal populations having the same mean?

Null hypothesis, $H_0 : \mu_A = \mu_B$

$$H_1 : \mu_A \neq \mu_B$$

Level of significance, $\alpha = 5\% = 0.05$

$$\text{Test statistic, } t = \frac{\bar{X} - \bar{Y}}{S \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

$$\bar{X} = \frac{\sum x_i}{n} \quad \& \quad \bar{Y} = \frac{\sum y_i}{n}$$

<u>x_i</u>	<u>$x_i - \bar{x}$</u>	<u>$(x_i - \bar{x})^2$</u>
24	-0.6	0.36
27	2.4	5.76
26	1.4	1.96
21	-3.6	12.96
25	0.4	0.16
		<u>21.04</u>

$$\bar{x} = \underline{\underline{24.6}}$$

<u>y_i</u>	<u>$y_i - \bar{y}$</u>	<u>$(y_i - \bar{y})^2$</u>
27	-2	4
30	1	1
28	-1	1
31	2	4
22	-7	49
36	7	49
		<u>108</u>

$$\bar{y} = \underline{\underline{29}}$$

$$S^2 = \frac{\sum (x_i - \bar{x})^2 + \sum (y_i - \bar{y})^2}{n_1 + n_2 - 2}$$

$$s^2 = \frac{21.04 + 108}{5+6-2} = \frac{129.04}{9}$$

$$= 14.33$$

$$s = \sqrt{14.33} = 3.78$$

$$t = \frac{\bar{X} - \bar{Y}}{s \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

$$\text{ie } t = \frac{24.6 - 29}{3.78 \sqrt{\frac{1}{5} + \frac{1}{6}}}$$

$$= \frac{-4.4}{2.2889} = -1.92.$$

$$|t| = 1.92.$$

From table, $t(0.05, 9) = 2.262.$

Calculated value of t ~~<~~ table value of t

$\therefore H_0$ is accepted.

ie, the two samples ^{came} from normal population having same mean.

2. A group of 5 patients treated with medicine A weighing 42, 39, 48, 60 and 41 kgs. Second group of 7 patients treated with medicine B weighing 38, 42, 56, 64, 68, 69 & 62 kgs. Do you agree with the claim that medicine B increases weight significantly?

$$H_0: \mu_A = \mu_B$$

$$H_1: \mu_A < \mu_B$$

Level of significance = 5% = 0.05

$$\text{Test statistic, } t = \frac{\bar{X} - \bar{Y}}{S \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

$$\bar{X} = 46, \bar{Y} = 57$$

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
42	-4	16
39	-7	49
48	2	4
60	14	196
41	-5	25
		290

$$\bar{X} = 46$$

y_i	$(y_i - \bar{y})$	$(y_i - \bar{y})^2$
38	-19	361
42	-15	225
56	-1	1
64	7	49
68	11	121
69	12	144
62	5	25
		<u>926</u>

$\bar{y} = \frac{\sum y_i}{n} = \underline{\underline{57}}$

$$s^2 = \frac{\sum (x_i - \bar{x})^2 + \sum (y_i - \bar{y})^2}{n_1 + n_2 - 2}$$

$$= \frac{290 + 926}{5 + 7 - 2} = 121.6$$

$$s = \sqrt{121.6} = \underline{\underline{11.02}}$$

$$t = \frac{\bar{x} - \bar{y}}{s \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{46 - 57}{11.02 \sqrt{\frac{1}{5} + \frac{1}{7}}}$$

$$= \frac{-11}{6.45265} = -1.704$$

$$|t| = 1.704.$$

Table value, $t(0.05, 10) = 1.812$ (one-tail)

Calculated value of $t <$ Table value of t .

$\therefore H_0$ is accepted

i.e., ~~S~~ Medicine B does not increase weight significantly
